# A brief overview of machine translation, neural networks and attention

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# Relevant Papers and Evolution

- Sequence to Sequence Learning with Neural Networks, Sutskever et al (2014): Encoder Decoder RNN for machine translation
- Neural Machine Translation by Jointly Learning to Align and Translate, Bahdanau et al (2014): Original attention mechanism
- Effective Approaches to Attention-based Neural Machine Translation, Luong et al (2015): Global vs local attention, input feeding
- Convolutional Sequence to Sequence Learning, Gehring et al (2017): Replaces RNN with CNN
- Attention Is All You Need, Vaswani et al (2017): All recurrence/convolution removed
- Focus was on translation, but paper notes potential applications for image, audio,

# Neural Network Basics

- Neurons
  - Weights applied to inputs
  - Summed and bias added
  - Activation function (relu, sigmoid)
- Layers
- Training
  - Labeled Training set (this picture is a cat/dog/etc)
  - Back propagation: seek minimums via gradient descent
- Standard Deep neural networks work with known-length

# Recurrent Neural Network

- Processing sequential data (order matters)
- Hidden state: piece of memory that is updated every cycle based on current input and previous hidden state value
- Vanishing gradient problem: (Extremely simplified) effects of earlier values become overshadowed by those later on, which may not be important
- Long short-term memory (LSTM): replaces multiplicative update with additive with option to ignore values

# Seq2seq and Encoder Decoder

- “The RNN can easily map sequences to sequences whenever the alignment between the inputs the outputs is known ahead of time. However, it is not clear how to apply an RNN to problems whose input and the output sequences have different lengths with complicated and non-monotonic relationships.” (Sutskever et al, p3) [1]
- Previous papers use two RNNs, one to read the input sequentially, and generate a fixed-dimension vector representation, which is then decoded by the second
- Adds deep LSTM to both RNNs, fixing long range dependency issue
- Reverse input but maintain standard order for output
- Mentions question answering (instead of from one language to another, maps from question to answer)

# Attention: “Learning to Align and Translate”

- When doing natural language tasks, how do you determine ordering of words
- Replaces the fixed vector output of the encoder with a sequence of vectors, that are selected from
- Bidirectional RNN Encoder: Forward and backwards RNN
  - Each word receives it’s annotation by concatenating the relevant hidden states from both
- RNN Decoder creates a context vector for each word: focuses on relevant words from the input. This is the attention mechanism
- Soft alignment: (softmax) Allows the model to consider relevant words, not just their ordering, but how they impact the word
- Significantly improves translating long sentences

# Effective Approaches: Global vs Local attention and input feeding

- Global approach attends all words, local only looks at a subset
- Global approach is similar to that used in original attention paper, Bahdanau et al. (2015)
- Local is computationally cheaper, outputs fixed dimension alignment vector
- Input Feeding: concatenate an additional vector to the input to keep information about past alignment

# Convolutional Neural Network

- Applies weights simultaneously across a matrix as a subset of a larger matrix (IE if convoluting over a 100x100 image, instead of applying weights to every pixel simultaneously, only apply weights to 10x10 at a time)
- Parallelization: computations not dependent on previous time steps
- Multi-layer CNNs
  - Nearby elements interact at lower layers, distant elements at higher layers
- Input element representation
  - Have to embed absolute position since we are no longer parsing sequentially
  - Same is done for output elements that are fed back into decoder network
- Issue is that window size of the CNN limits words from the beginning and end being compared

# Transformer Architecture

- Bypass sequential nature of recurrent implementation of attention
- Unlike CNN approach, relation between distant words can still be accounted for
- Giant word embedding table: each word gets its own row (say 50,000 words x 512 columns)
- Position embedding table: added to the word vector (512x512)
  - Number of rows determines maximum sequence size. Column length matches word embedding table
- Parallel encoder and decoder stacks
- After encoding each word + position entry, it gets fed into every decoder